DATA CLEANING AND VISUALIZATION USING PYTHON

1. Introduction

Data cleaning and visualization are important steps in data analysis. Raw datasets often contain missing values, duplicate records, and unusual values that can affect the accuracy of analysis.

This project uses Python to clean and analyze a sample sales dataset. The cleaned data is visualized using charts to identify sales patterns and compare product performance.

2. Objectives

The main objectives of this project are:

- To explore and understand a raw dataset.
- To identify and handle missing values.
- To detect and remove duplicate records.
- To identify potential outliers.
- To calculate important sales statistics.
- To visualize sales data using graphs.
- To communicate findings through data analysis.

3. Tools and Technologies

The following tools and libraries were used:

- Python: Programming language.
- Jupyter Notebook: Development environment.
- Pandas: Data cleaning and manipulation.
- NumPy: Numerical operations.
- Matplotlib: Data visualization.
- Seaborn: Statistical visualization.

4. Dataset Description

A sample sales dataset was created for this project.

The dataset contains the following columns:

- Order_ID: Unique identifier assigned to an order.
- Product: Name of the product sold.
- Category: Product category.
- Quantity: Number of units sold.
- Price: Price per unit.
- Sales: Total sales value for the order.

The original dataset contained 10 rows and 6 columns.

5. Data Cleaning

Several data-cleaning operations were performed.

5.1 Handling Missing Values

One missing value was identified in the Quantity column. It was replaced using the median quantity.

5.2 Removing Duplicates

One duplicate record was identified and removed using Pandas.

5.3 Outlier Detection

The Interquartile Range (IQR) method was used to identify potential outliers in the Sales column. Potential outliers were evaluated rather than automatically deleted.

5.4 Verification

After cleaning, the dataset contained 9 rows and 6 columns, with no missing values or duplicate rows.

6. Data Analysis

The cleaned dataset was analyzed to calculate:

- Total sales.
- Average sales per record.
- Highest and lowest sales values.
- Total quantity sold.
- Sales generated by each product.

These calculations helped compare the sales performance of different products.

7. Data Visualization

Four types of visualizations were created.

7.1 Bar Chart

The bar chart compares total sales for laptops, phones, and tablets. It helps identify which product generated the most revenue.

7.2 Pie Chart

The pie chart displays each product's contribution to total sales as a percentage.

7.3 Histogram

The histogram illustrates the distribution of sales values and shows how frequently different sales amounts occur.

7.4 Box Plot

The box plot displays the spread of sales values and helps identify potential outliers.

8. Key Findings

Based on the sample dataset:

- Laptops generated the highest total sales.
- Phones generated the second-highest total sales.
- Tablets generated the lowest total sales.
- Data cleaning reduced the dataset from 10 rows to 9 rows.
- The missing quantity was filled, and the duplicate record was removed.
- Visualizations made it easier to compare products and understand the sales distribution.

9. Conclusion

This project demonstrated how Python can be used to clean, analyze, and visualize data. Pandas helped identify missing values and remove duplicate records, while Matplotlib and Seaborn were used to create visualizations.

The analysis showed how charts can reveal patterns in product sales and support data-driven decisions.

The project provided practical experience in data preprocessing, statistical analysis, visualization, and communicating insights from data.

10. Future Improvements

The project could be improved by using a larger real-world dataset, analyzing monthly sales trends, comparing more product categories, and creating an interactive dashboard.